

# magicpin: Connecting offline retailers with customers through data- rich experiences

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magicpin connects  
offline retailers and customers through a  
digital platform to  
enhance the customer experience  
and increase brand  
visibility.

## About magicpin

magicpin (<https://magicpin.in/>) is an online location intelligence platform that allows customers to discover retailers in their vicinity and get discounts at them. Its 300,000 retail merchants use the platform to engage with or provide personalized offers to their customers. magicpin is the number one retail discovery app in India with over 10 million users, and is operational in 12 major Indian cities.

Industries: Retail & Consumer Goods

## Google Cloud results

- Scales easily to handle increased monthly transactions from 100,000 to more than 3 million
- Cuts reporting time from 3 days to just minutes
- Onboards more than 300,000 retailers onto its platform with ease
- Helps to support goal of reaching 100% of offline retailers

Helps  
300,000  
gain new  
visibility

Location: India

Cloud SQL (<https://cloud.google.com/sql/>),



## About Searce

Searce is a technology consulting firm that empowers clients to future-proof their businesses by leveraging cloud, AI technologies and more. Based in India, Searce is an award-winning Google Cloud partner

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[Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>)

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[BigQuery](https://cloud.google.com/bigquery/) (<https://cloud.google.com/bigquery/>)

[Cloud Bigtable](https://cloud.google.com/bigtable/) (<https://cloud.google.com/bigtable/>)

[Cloud CDN](https://cloud.google.com/cdn/) (<https://cloud.google.com/cdn/>)

[Cloud DNS](https://cloud.google.com/dns/) (<https://cloud.google.com/dns/>)

[Cloud SQL](https://cloud.google.com/sql/) (<https://cloud.google.com/sql/>)

[Compute Engine](https://cloud.google.com/compute/)

(<https://cloud.google.com/compute/>)

[Dataflow](https://cloud.google.com/dataflow/) (<https://cloud.google.com/dataflow/>)

[Looker Studio](https://cloud.google.com/looker-studio/)

(<https://cloud.google.com/looker-studio/>)

[Kubernetes Engine](https://cloud.google.com/kubernetes-engine/)

(<https://cloud.google.com/kubernetes-engine/>)

Retailers play an integral part in reality is that most of these business dark, with many of them working connected world that still relies of customer engagement.

Vertex AI (<https://cloud.google.com/vertex-ai/>)  
partner

Vision AI (<https://cloud.google.com/vision/>)

Cloud Security Command Center  
(<https://cloud.google.com/security-command-center/>)

This is something that [magicpin](https://magicpin.in/) (<https://magicpin.in/>), an Indian online location intelligence platform, seeks to improve in retail verticals like food, fashion, pharma, and electronics. With over 10 million users on its platform covering 12 major Indian cities and growing, it has firmly placed a stake as the number one retail discovery app in India.

"If we can make the offline space data-rich, we can enable users to discover retail stores, visit them, and transact. We can also enable these stores to reach out to the user, and that's a big service to both parties. If you think about it, even in the best economies, 60% to 80% of retail sales are still offline. Bridging the gap between retailers and consumers is where the magic is," explains Pranay Gupta, Vice President of Growth and Business Development at magicpin.



## Helping small retailers reach bigger audiences

Pranay believes that digitizing local retail stores will reshape their customers' perception of what's possible. With many of these retailers being one- or two-owner setups, they are unable to dedicate resources to running their own marketing team to improve customer engagement.

magicpin's platform offers a marketing solution that bridges this gap for retailers. It helps retailers interact with customers, manage transactions easily, ensure personalized communications to users, and more. The solution also shows tailor-made offers based on a customer's preferences inferred from their transaction history with a store.

As a result, offline retailers can engage in performance marketing, and only pay a charge after they have realized revenue. Customers on the other hand, are incentivized to constantly discover more, because the platform shows new discounts and enables loyalty points collection when customers upload receipts.

These improvements to the retail experience for both retailers and customers have persuaded over 300,000 retailers to onboard with magicpin's platform.

"We realized that scaling would require more and more database staff and domain experts. That's when we decided that we should move to Cloud Bigtable as soon as possible."

*—Kunal Gupta, Vice  
President of Technology,  
magicpin*

## **Scaling fast to help thousands of offline retailers**

magicpin has been working closely with Google since its early days when it was selected to be part of Google's Launchpad program (now called Accelerator (<https://startup.google.com/accelerator/>)). Google teams from Firebase, Google Merchant Center, Google Maps and Google Play worked closely with magicpin to support it in its growth journey. As an alumnus, magicpin received early

exposure to [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>), which revealed solutions to many of magicpin's functions across cataloging, search, payments, and delivery.

This heavily contrasted with a legacy database solution that could not keep pace with the startup's rapid growth in 2016. Integration with Google Cloud quickly resolved the "panic mode" magicpin found itself in after its rapid growth caused database crashes.

"We realized that scaling would require more and more database staff and domain experts. That's when we decided that we should move to [Cloud Bigtable](https://cloud.google.com/bigtable) (<https://cloud.google.com/bigtable>) as soon as possible," shares Kunal Gupta, Vice President Of Technology at magicpin.

After integrating with Bigtable, they were also free to scale as much as they wanted, enabling it to handle a 400% increase in monthly transactions, from 10,000 to 40,000. Integration with a NoSQL database service has also been deemed "phenomenal" for the team in other ways, because magicpin didn't need to hire any experts in database management.

This was key to a startup that was concerned about staffing and infrastructure spend. With various cost optimization and performance levers such as committed use discounts (CUDs), high-performance archival storage access, and AMD-based instances, they have been able to save more than 30% in annual infrastructure costs.

"At every point in time, Google has been supportive of our needs, and has always given us nudges on how we could be thinking about some of our applications, whether it be re-architecting them or applying some other discounts, like CUDs, to save on costs," Kunal explained.

Another key to magicpin's success is support from Google Cloud Partner, Searce, who has been by the startup's side since the start of the migration. Searce first started by designing a target state diagram for a Lift and Shift Migration to Google Cloud, and in three years, the company has grown from being a migration partner, to a full-fledged DevOps-first partner.

Today, Searce is helping magicpin with key initiatives such as building CI/CD pipelines and implementing security best practices. It also ensures that magicpin's infrastructure is error free, deploying quarterly health checks and providing security reports as well.

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*—Kunal Gupta, Vice  
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## **Making magic with more cloud solutions**

The considerable computing behind magicpin is handled by Compute Engine

(<https://cloud.google.com/compute>) and Google

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE). The

team discovered that GKE provided the right

abstraction not just for scale management, but also

automatically handled service availability. The cherry

on top, adds Kunal, is that GKE is more cost-efficient

for its use case.



"This is why we've undertaken a major exercise to move 100% of our workload onto GKE by the end of 2022. We are confident to bet on infrastructure evolution towards more containerized and lightweight operations with GKE," Kunal explains.

Where magicpin's marketing and product teams used to spend days moving data from its previous Online Transaction Processing (OLTP) to Online Analytical Processing (OLAP) systems, it now only takes minutes with Cloud SQL

(<https://cloud.google.com/sql>) and BigQuery

(<https://cloud.google.com/bigquery>). The faster data flow streams across magicpin's modern OLTP and OLAP systems speed up user profile creation, user segmentation, and the creation of tailored experiences for each segment.

"Before BigQuery, our analytics have been drastically different. I remember there were days when I would work with analysts on reporting, and it could take up to three days sometimes because they will run a really long query. After it runs through the night, we would realize that something had gone haywire. BigQuery has made reporting much more efficient for us by cutting reporting times to just minutes."

"Also, our recent experience with Looker Studio

(<https://cloud.google.com/looker-studio>) has replaced a lot of the code-based analytics that we were doing. All our senior managers have to do now is drag and drop to modify reports as and when needed and they can also play around with various input parameters to see how reports are changing," Pranay says.

As the amount and quality of data grows on magicpin's platform, it has embarked on leveraging powering machine learning capabilities to improve its customer experience.

"We are also ramping up our use of Vertex AI (<https://cloud.google.com/vertex-ai>) for machine learning. We have so many ideas because we now have so much valuable data. We are heavily using Vertex AI to build more personalized and tailored experiences for our users," says Kunal.

magicpin's engineers use up to 35 user features to train its machine learning models, including features like a user's age, gender, past transactions, locality, or recent searches.

These features are used to create a profile score that influences experience personalization, product margin contribution, or premium support levels. A low score is used to inform which users' transactions should be limited to prevent fraud.

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We envision

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India GEO team  
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the Google  
Partner program  
and Google  
Maps services  
to deliver more  
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*—Pranay Gupta, Vice  
President of Growth and  
Development, magicpin*

## **Towards a more data-rich retail future**

"The main advantage in the ecommerce space is the speed of service delivery and getting instant gratification. That's why we have to learn and iterate fast to beat the competition. The services that Google provides make our job very simple. Our business team comes up with the business problems, our tech team focuses on solving those problems, and Google Cloud takes care of our infrastructure," Kunal reveals.

The support offered to magicpin adds to its efforts to improve its penetration into the Indian market, and its goal to cover 100% of offline retailers. Google

Cloud's offerings, when paired with Google's search capabilities, are paving the way for more possibilities.

"Google Cloud provides a strategic foundation on which we can really propel growth," says Pranay.

"What magicpin has been able to do amazingly well is get verified access to local stores. We have their location, data, inventory, and catalog information.

We envision working with the India GEO team to do more with the [Google Partner](#)

(<https://www.google.com/partners/become-a-partner/>) program and [Google Maps](#)

(<https://www.google.com/maps/about/#!/>) services to deliver more visibility for local retail stores," he concludes.